

REAL-WORLD CONSERVATION LAWS · DENNIS G. ZILL & COURSE SYLLABUS W4

Chapters 2.7 & 2.8: First-Order Engineering Modeling

Formulating differential equations for brine mixing tanks, Newton's law of cooling, Torricelli's tank draining, and logistic growth.

Zill §2.7–2.8 (pp. 109–130)

Syllabus Week 4 Core

Applied Engineering Exam Focus

PLAIN-ENGLISH INTUITION

Engineering modeling boils down to one universal truth: Rate of Accumulation = Input Rate – Output Rate. Setting up the balance equation requires careful physical unit tracking. Once formulated, every model resolves into either a separable ODE or a first-order linear ODE!

1. Newton's Law of Cooling / Warming

Rate of temperature change is proportional to difference with surroundings T_m :

$$\frac{dT}{dt} = k(T - T_m) \quad (k < 0)$$

Explicit Solution: $T(t) = T_m + (T_0 - T_m)e^{kt}$. As $t \rightarrow \infty$, $T(t) \rightarrow T_m$ asymptotically.

2. Tank Mixing / Solute Dilution

Accumulation of solute mass $A(t)$ in a well-stirred liquid volume $V(t)$:

$$\frac{dA}{dt} = c_{\text{in}}r_{\text{in}} - \left(\frac{A(t)}{V(t)}\right)r_{\text{out}}$$

- Volume: $V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t$.
- If $r_{\text{in}} \neq r_{\text{out}}$, volume changes $\implies$ **Linear ODE!**

Master Worked Example 1 • Variable-Volume Tank (Zill p. 115)

MIDTERM FAVORITE

Problem: A 300-gallon tank initially holds 100 gal of pure water. Brine with 2 lb/gal salt enters at 3 gal/min, and the stirred mixture exits at 2 gal/min. Find salt $A(t)$ until the tank overflows at 300 gallons.

1 Volume & Overflow Time:

$$V(t) = 100 + (3 - 2)t = 100 + t \text{ gal.}$$

$$\text{Full when } V(t) = 300 \implies t = 200 \text{ min.}$$

2 Formulate Linear ODE:

$$R_{\text{in}} = (2 \text{ lb/gal})(3 \text{ gal/min}) = 6 \text{ lb/min.}$$

$$R_{\text{out}} = \frac{A(t)}{100+t} \times 2 = \frac{2}{100+t}A(t).$$

$$\frac{dA}{dt} + \frac{2}{100+t}A = 6$$

3 Solve via Integrating Factor:

$$\mu(t) = e^{\int \frac{2}{100+t} dt} = (100+t)^2.$$

$$\frac{d}{dt}[A(100+t)^2] = 6(100+t)^2 \implies A(100+t)^2 = 2(100+t)^3 + C.$$

$$A(t) = 2(100+t) + \frac{C}{(100+t)^2}.$$

$$\text{Since } A(0) = 0 \implies 0 = 200 + \frac{C}{10000} \implies C = -2 \times 10^6.$$

$$A(t) = 2(100+t) - \frac{2 \times 10^6}{(100+t)^2}$$

$$\text{At } t = 200: A(200) = 2(300) - \frac{2 \times 10^6}{90000} = 577.8 \text{ lbs.}$$

2 Nonlinear Modeling: Torricelli's Draining Law & Logistic Growth

3. Torricelli's Law (Draining Tanks)

Water height $h(t)$ leaking through a hole of cross-sectional area a at the bottom of a tank of area $A(h)$:

$$A(h) \frac{dh}{dt} = -ac\sqrt{2gh}$$

- c : discharge coefficient ($0 < c \leq 1$).
- g : gravity (32 ft/s^2 or 9.8 m/s^2).
- Solved by **separation of variables**!

4. Logistic Population Growth

Population $P(t)$ with finite carrying capacity K :

$$\frac{dP}{dt} = P(a - bP) = aP \left(1 - \frac{P}{K}\right)$$

- Carrying capacity: $K = \frac{a}{b}$.
- Critical points: $P = 0$ (unstable repeller) and $P = K$ (stable attractor).

Master Worked Example 2 • Torricelli's Draining Tank (Zill p. 122)

NONLINEAR SEPARATION

Problem: A cylindrical tank of radius $R = 2 \text{ ft}$ and height $H = 9 \text{ ft}$ is full of water. A circular hole of radius $r = 1 \text{ in} = \frac{1}{12} \text{ ft}$ at the bottom is opened at $t = 0$. Using $g = 32 \text{ ft/s}^2$ and discharge coefficient $c = 1$, find $h(t)$ and determine how long it takes to drain completely.

1 Compute Cross-Sectional Areas:

$$\text{Tank area: } A_h = \pi R^2 = \pi(2)^2 = 4\pi \text{ ft}^2.$$

$$\text{Hole area: } a = \pi r^2 = \pi \left(\frac{1}{12}\right)^2 = \frac{\pi}{144} \text{ ft}^2.$$

$$\text{Velocity: } \sqrt{2gh} = \sqrt{2(32)h} = 8\sqrt{h} \text{ ft/s.}$$

2 Torricelli DE:

$$4\pi \frac{dh}{dt} = -\frac{\pi}{144}(8\sqrt{h}) \implies \frac{dh}{dt} = -\frac{1}{72}\sqrt{h}.$$

3 Separate Variables & Integrate:

$$\frac{dh}{\sqrt{h}} = -\frac{1}{72}dt \implies 2\sqrt{h} = -\frac{1}{72}t + C_1.$$

$$\text{At } t = 0, h = 9 \implies 2\sqrt{9} = C_1 \implies C_1 = 6.$$

$$2\sqrt{h} = 6 - \frac{t}{72} \implies \sqrt{h} = 3 - \frac{t}{144}.$$

$$h(t) = \left(3 - \frac{t}{144}\right)^2$$

4 Time to Completely Drain ($h = 0$):

$$3 - \frac{t}{144} = 0 \implies t = 432 \text{ seconds} = 7.2 \text{ minutes.}$$

THE UNIT CONVERSION TRAP IN TORRICELLI'S LAW

Always convert the hole radius into feet if gravity is in ft/s^2 ($g = 32$)! Using $r = 1 \text{ inch}$ directly with $g = 32$ produces an answer that is off by a factor of 144!